RET fusion-hotspot benchmark report

Study: thca_tcga_pan_can_atlas_2018

Abstract

This report analyzes RET gene fusions from the thca_tcga_pan_can_atlas_2018 study, covering 36 fusion events. 35/35 fusions with known frame status (100.0%) were in-frame. 33/36 fusions (91.7%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=1, not statistically significant at alpha=0.05). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 100.0% in-frame and 91.7% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 36 analyzed fusion events, identifying 10 distinct fusion partner genes. The most frequent partner was CCDC6, observed in 22 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 33/35 (94.3%) of in-frame fusions retained the domain, p=1 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.001998.

Domain disruption

Disruption of the Cadherin domain was tested with Fisher's exact test comparing in-frame fusions against all others; 32/33 (97.0%) of in-frame fusions disrupted the domain, p=1 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Junction concentration

97% of 32 fusions with this gene as the 3' partner join at exon 12, against 4% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.000999001, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 33 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 712 aa (exon 11; label-permutation p=0.0659341). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -13 aa from the nearest configured domain boundary at 725 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=32) and "not_retained" (n=1). For Frame_status, retained: 32/32 (100.0%, 95% CI 89.3%-100.0%); not_retained: 1/1 (100.0%, 95% CI 20.7%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of unavailable vs unavailable, p=unavailable.

Composite evidence score

Composite evidence score reported: cutpoint_proximity sub-score excluded: corrected cutpoint p-value was missing or >= 0.05, cutpoint was not determinable, or mapped breakpoints were unavailable.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: RET: fusion-positive-vs-negative expression comparison skipped; no sample had both established SV assay coverage and measured expression, so no fusion-negative group could be defined.

Genomic position recurrence

Genomic position recurrence produced results for this run (2 row(s) in genomic_bin_recurrence, 1 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: RET has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: RET has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection reported: no candidate window of the configured widths (25, 50, 100, 200) met the minimum-event-count guard (min_events_per_window=4) on both sides of the split

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CCDC6	22	0.6111111111111112	0.26994040818153375	0.30004340774793187	None	0.5496850580610834	confidence_certainty, domain_disruption, domain_retention, recurrence	0.43034779145224883	1
NCOA4	5	0.1388888888888889	0.26994040818153375	0.30004340774793187	None	0.5496850580610834	confidence_certainty, domain_disruption, domain_retention, recurrence	0.26368112478558214	2
ERC1	2	0.05555555555555555	0.26994040818153375	0.30004340774793187	None	0.5496850580610834	confidence_certainty, domain_disruption, domain_retention, recurrence	0.2342693600796998	3
AKAP13	1	0.027777777777777776	0.26994040818153375	0.30004340774793187	None	0.5496850580610834	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22446543851107237	4
DLG5	1	0.027777777777777776	0.26994040818153375	0.30004340774793187	None	0.5496850580610834	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22446543851107237	5
FKBP15	1	0.027777777777777776	0.26994040818153375	0.30004340774793187	None	0.5496850580610834	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22446543851107237	6
MRLN	1	0.027777777777777776	0.26994040818153375	0.30004340774793187	None	0.5496850580610834	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22446543851107237	7
SPECC1L	1	0.027777777777777776	0.26994040818153375	0.30004340774793187	None	0.5496850580610834	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22446543851107237	8
TBL1XR1	1	0.027777777777777776	0.26994040818153375	0.30004340774793187	None	0.5496850580610834	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22446543851107237	9
TRIM27	1	0.027777777777777776	0.26994040818153375	0.30004340774793187	None	0.5496850580610834	confidence_certainty, domain_disruption, domain_retention, recurrence	0.22446543851107237	10

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
663	1	31	0	1	None	1.0	-0.0
712	1	31	1	0	0.0	0.06060606060606061	1.2174839442139063

domain_disruption tables

frame_domain_contingency_table

32	0
1	0

frame_domain_contingency_tables_by_tumor_type

stratum	table
THPA	[32, 0], [1, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
cadherin	Cadherin domain	33	1	0	32	0.0

domain_retention tables

frame_domain_contingency_table

32	0
1	0

frame_domain_contingency_tables_by_tumor_type

stratum	table
THPA	[32, 0], [1, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	33	32	0	1	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
12	32	33	0.9696969696969697

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
AKAP13	1	1	1	1
CCDC6	22	22	21	21
DLG5	1	1	1	1
ERC1	2	2	2	2
FKBP15	1	1	1	1
MRLN	1	1	1	1
NCOA4	5	5	4	4
SPECC1L	1	1	1	1
TBL1XR1	1	1	1	1
TRIM27	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
10	43116000	43117000	31	EVT-TCGA-BJ-A0ZJ-01-2, EVT-TCGA-BJ-A28Z-01-22, EVT-TCGA-CE-A13K-01-20, EVT-TCGA-CE-A481-01-36, EVT-TCGA-CE-A482-01-30, ... (31 total)
10	43114000	43115000	2	EVT-TCGA-DJ-A4V5-01-31, EVT-TCGA-EL-A3ZP-01-13

exact_position_recurrence

chromosome	position	n_events	event_ids
10	43116584	31	EVT-TCGA-BJ-A0ZJ-01-2, EVT-TCGA-BJ-A28Z-01-22, EVT-TCGA-CE-A13K-01-20, EVT-TCGA-CE-A481-01-36, EVT-TCGA-CE-A482-01-30, ... (31 total)

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
713	31	1	0	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	12	31	32	0.96875	0.03741209563994374	42.747509348706295
five_prime	11	1	1	1.0	0.03945616502578528	1.403885128639765
three_prime	11	1	32	0.03125	0.019896858884200657	0.3239014554305676

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-FK-A3SG-01-1	TCGA-FK-A3SG-01	TCGA-FK-A3SG	Thyroid Cancer	THPA	AKAP13-RET	AKAP13	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | AKAP13-RET Fusion	in-frame
EVT-TCGA-BJ-A0ZJ-01-2	TCGA-BJ-A0ZJ-01	TCGA-BJ-A0ZJ	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-J8-A4HW-01-3	TCGA-J8-A4HW-01	TCGA-J8-A4HW	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-DJ-A2Q1-01-4	TCGA-DJ-A2Q1-01	TCGA-DJ-A2Q1	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-DJ-A3V3-01-5	TCGA-DJ-A3V3-01	TCGA-DJ-A3V3	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-DJ-A4UQ-01-6	TCGA-DJ-A4UQ-01	TCGA-DJ-A4UQ	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-DJ-A4V5-01-7	TCGA-DJ-A4V5-01	TCGA-DJ-A4V5	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-DO-A1JZ-01-8	TCGA-DO-A1JZ-01	TCGA-DO-A1JZ	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-E3-A3E0-01-9	TCGA-E3-A3E0-01	TCGA-E3-A3E0	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-E8-A44M-01-10	TCGA-E8-A44M-01	TCGA-E8-A44M	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-EL-A3CY-01-11	TCGA-EL-A3CY-01	TCGA-EL-A3CY	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-EL-A3TB-01-12	TCGA-EL-A3TB-01	TCGA-EL-A3TB	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-EL-A3ZP-01-13	TCGA-EL-A3ZP-01	TCGA-EL-A3ZP	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43114587	11	663	False	663	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-EL-A3ZS-01-14	TCGA-EL-A3ZS-01	TCGA-EL-A3ZS	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-EM-A3AN-01-15	TCGA-EM-A3AN-01	TCGA-EM-A3AN	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-ET-A39R-01-16	TCGA-ET-A39R-01	TCGA-ET-A39R	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-ET-A3DR-01-17	TCGA-ET-A3DR-01	TCGA-ET-A3DR	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-ET-A40T-01-18	TCGA-ET-A40T-01	TCGA-ET-A40T	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-FK-A3SE-01-19	TCGA-FK-A3SE-01	TCGA-FK-A3SE	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-CE-A13K-01-20	TCGA-CE-A13K-01	TCGA-CE-A13K	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-FK-A4UB-01-21	TCGA-FK-A4UB-01	TCGA-FK-A4UB	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame
EVT-TCGA-BJ-A28Z-01-22	TCGA-BJ-A28Z-01	TCGA-BJ-A28Z	Thyroid Cancer	THPA	CCDC6-RET	CCDC6	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | CCDC6-RET Fusion	in-frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-DE-A69K-01-23	TCGA-DE-A69K-01	TCGA-DE-A69K	Thyroid Cancer	THPA	DLG5-RET	DLG5	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | DLG5-RET Fusion	in-frame
EVT-TCGA-EL-A3T9-01-24	TCGA-EL-A3T9-01	TCGA-EL-A3T9	Thyroid Cancer	THPA	ERC1-RET	ERC1	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | ERC1-RET Fusion	in-frame
EVT-TCGA-EM-A3FQ-01-25	TCGA-EM-A3FQ-01	TCGA-EM-A3FQ	Thyroid Cancer	THPA	ERC1-RET	ERC1	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | ERC1-RET Fusion	in-frame
EVT-TCGA-ET-A3DQ-01-26	TCGA-ET-A3DQ-01	TCGA-ET-A3DQ	Thyroid Cancer	THPA	FKBP15-RET	FKBP15	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | FKBP15-RET Fusion	in-frame
EVT-TCGA-EL-A3H3-01-27	TCGA-EL-A3H3-01	TCGA-EL-A3H3	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | NCOA4-RET Fusion	in-frame
EVT-TCGA-EM-A2CU-01-28	TCGA-EM-A2CU-01	TCGA-EM-A2CU	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | NCOA4-RET Fusion	in-frame
EVT-TCGA-DE-A3KN-01-29	TCGA-DE-A3KN-01	TCGA-DE-A3KN	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False												Protein kinase domain	Cadherin domain		NA | NCOA4-RET Fusion	in-frame

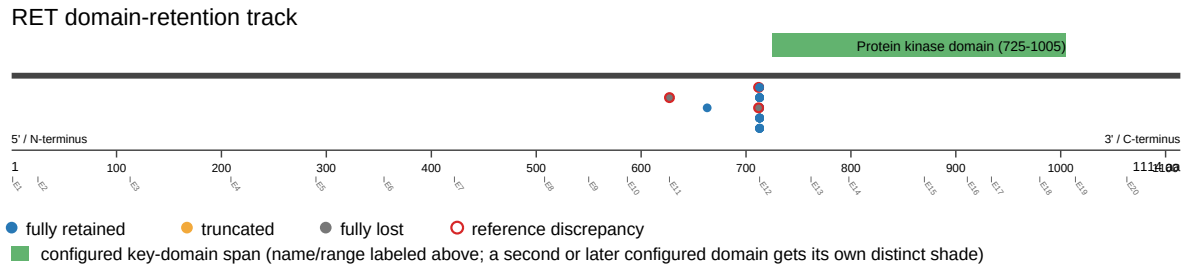
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-CE-A482-01-30	TCGA-CE-A482-01	TCGA-CE-A482	Thyroid Cancer	THPA	NCOA4-RET	NCOA4	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | NCOA4-RET Fusion	in-frame
EVT-TCGA-DJ-A4V5-01-31	TCGA-DJ-A4V5-01	TCGA-DJ-A4V5	Thyroid Cancer	THPA	RET-CCDC6	CCDC6	in-frame	five_prime	43114736	11	712	False	712	True	exact match	lost	0.0	False													Cadherin domain	Protein kinase domain		NA | RET-CCDC6 Fusion	in-frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-CE-A13K-01-32	TCGA-CE-A13K-01	TCGA-CE-A13K	Thyroid Cancer	THPA	RET-MRLN	MRLN	unknown	five_prime	43113675	10	627	False	626	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False														Cadherin domain	Protein kinase domain		NA | RET-MRLN Fusion	NA
EVT-TCGA-CE-A482-01-33	TCGA-CE-A482-01	TCGA-CE-A482	Thyroid Cancer	THPA	RET-NCOA4	NCOA4	in-frame	five_prime	43114736	11	712	False	712	True	exact match	lost	0.0	False												Cadherin domain	Protein kinase domain		NA | RET-NCOA4 Fusion	in-frame

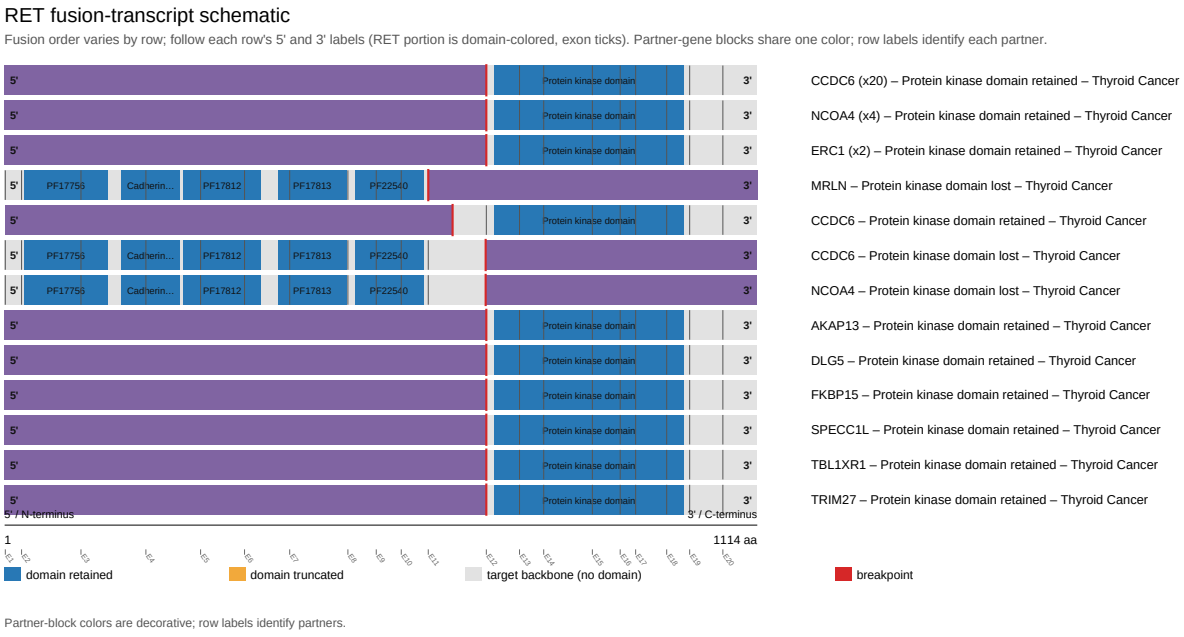
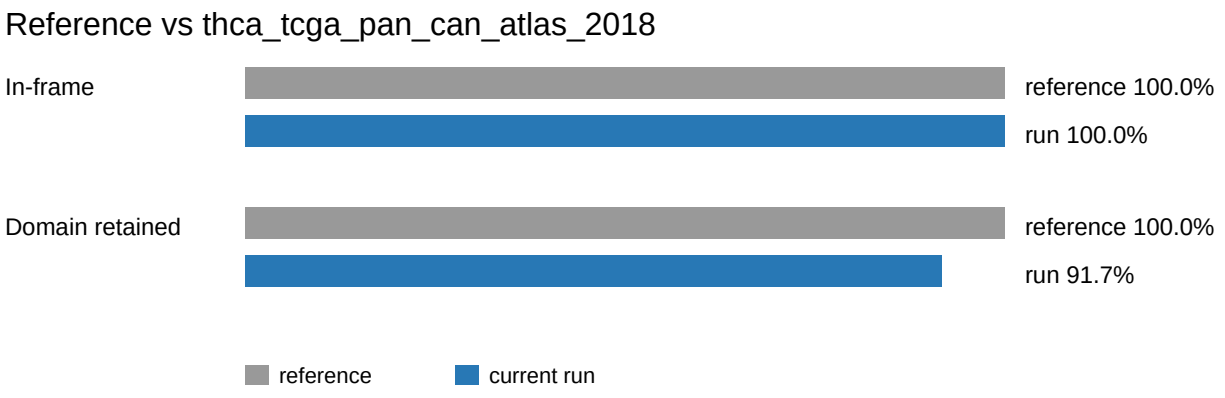
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-CE-A485-01-34	TCGA-CE-A485-01	TCGA-CE-A485	Thyroid Cancer	THPA	SPECC1L-RET	SPECC1L	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | SPECC1L-RET Fusion	in-frame
EVT-TCGA-ET-A40R-01-35	TCGA-ET-A40R-01	TCGA-ET-A40R	Thyroid Cancer	THPA	TBL1XR1-RET	TBL1XR1	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | TBL1XR1-RET Fusion	in-frame
EVT-TCGA-CE-A481-01-36	TCGA-CE-A481-01	TCGA-CE-A481	Thyroid Cancer	THPA	TRIM27-RET	TRIM27	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False													Protein kinase domain	Cadherin domain		NA | TRIM27-RET Fusion	in-frame

Figures

domain retention outliers



fusion schematic



reference comparison